

Alumno: Samuel Carreto Barrientos

Grupo: 3

IP asignada /24: 192.168.68.50/24

Gateway: 192.168.68.1

DNS: 8.8.8.8

Interfaz usada: enp0s3

Evidencias

1. ip -4 addr y ip route:

```
sam@sam03:~$ ip -4 addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    inet 192.168.68.50/24 brd 192.168.68.255 scope global enp0s3
        valid_lft forever preferred_lft forever
sam@sam03:~$ ip route
default via 192.168.68.1 dev enp0s3 proto static
192.168.68.0/24 dev enp0s3 proto kernel scope link src 192.168.68.50
```

2. resolvectl status y resolvectl query ubuntu.com:

```
sam@sam03:~$ resolvectl status
Global
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    resolv.conf mode: stub

Link 2 (enp0s3)
    Current Scopes: DNS
    Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    Current DNS Server: 8.8.8.8
    DNS Servers: 8.8.8.8 1.1.1.1
sam@sam03:~$ resolve query ubuntu.com
resolve: command not found
sam@sam03:~$ resolvectl status
Global
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    resolv.conf mode: stub

Link 2 (enp0s3)
    Current Scopes: DNS
    Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    Current DNS Server: 8.8.8.8
    DNS Servers: 8.8.8.8 1.1.1.1
sam@sam03:~$ resolvectl query ubuntu.com
ubuntu.com: 185.125.190.21 -- link: enp0s3
             185.125.190.29 -- link: enp0s3
             185.125.190.20 -- link: enp0s3
             2620:2d:4000:1::27 -- link: enp0s3
             2620:2d:4000:1::28 -- link: enp0s3
             2620:2d:4000:1::26 -- link: enp0s3

-- Information acquired via protocol DNS in 62.5ms.
-- Data is authenticated: no; Data was acquired via local or encrypted transport: no
-- Data from: network
```

3. systemctl status ssh:

```
sam@sam03:~$ systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-11-11 05:46:53 UTC; 46min ago
   TriggeredBy: • ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 1846 (sshd)
     Tasks: 1 (limit: 2268)
    Memory: 1.2M (peak: 1.5M)
       CPU: 26ms
    CGroup: /system.slice/ssh.service
            └─1846 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Nov 11 05:46:53 sam03 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Nov 11 05:46:53 sam03 sshd[1846]: Server listening on 0.0.0.0 port 22.
Nov 11 05:46:53 sam03 sshd[1846]: Server listening on :: port 22.
Nov 11 05:46:53 sam03 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

4. ufw status numbered:

5. Pruebas del servicio (HTTP headers, psql, etc.):

```
sam@sam03:~$ sudo ufw status numbered
Status: active

    To Action From
    --
[ 1] 22/tcp ALLOW IN 192.168.68.0/24
[ 2] 22/tcp LIMIT IN Anywhere
[ 3] 80/tcp ALLOW IN Anywhere
[ 4] 22/tcp (v6) LIMIT IN Anywhere (v6)
[ 5] 80/tcp (v6) ALLOW IN Anywhere (v6)
```

Reflexión (5-8 líneas)

Dejé abierto el puerto 22/tcp para la administración segura por SSH, limitado a la subred del laboratorio, y los puertos 80/tcp y 443/tcp para el servicio web de Nginx. Esta configuración hace que solo los servicios esenciales estén disponibles desde la red, mejorando la seguridad de la red y evitando accesos no autorizados a esta misma.

Referencias Netplan (Ubuntu networking): <https://netplan.io/examples/> / <https://netplan.io/reference>

systemd-resolved / resolvectl:

<https://manpages.ubuntu.com/manpages/noble/en/man1/resolvectl.1.html>

OpenSSH (servidor y configuración): <https://www.openssh.com/manual.html> / man
sshd_config

UFW (Ubuntu Firewall): <https://help.ubuntu.com/community/UFW>

Nginx en Ubuntu: <https://ubuntu.com/tutorials/install-and-configure-nginx>